

Local Spectral Envelopes for ADMM Registration: Exact Algebra, a Certificate, and a Negative Parameter-Selection Result

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Abstract

We investigate whether local Fourier envelopes can select ADMM penalty and relaxation parameters for quadratic subproblems in diffeomorphic registration without numerical hyperparameter search. For the precise scaled over-relaxation convention, we prove that the nonzero state spectrum equals that of $(1 - \alpha/2)I + (\alpha/2)C_H C_G$. Frozen constant-coefficient patches therefore have the exact modal factor $1 - \alpha + \alpha(\rho^2 + hg)/((\rho + h)(\rho + g))$, whose worst value occurs at curvature corners. A resolvent argument gives the noncommuting certificate $\varrho(T) \leq U_{\text{LFA}} + \alpha(\delta_H + \delta_G)/\rho$, and standard-ADMM minimization reduces to finitely many algebraic roots. Double-precision tests confirm the identities and certificate. Constant periodic problems agree with the local envelope to at most 1.0×10^{-15} in our three tested grids. The central practical hypothesis, however, fails in a predeclared controlled experiment: for all 18 variable-coefficient cases the certified joint selector chooses the lower relaxation boundary ($\alpha = 10^{-6}$) and upper penalty boundary ($\rho = 100$), giving convergence factors essentially one; its median relative gap to direct joint spectral optimization is 2.904. The certificate remains valid but is too conservative for tuning. We therefore report a negative result and stop before real-data registration under the predeclared decision gate. The algebra, code, configurations, raw results, and failure analysis are released as a reproducible basis for tighter structured perturbation bounds.

1 Introduction

Diffeomorphic image registration estimates an invertible, topology-preserving spatial transformation that aligns a moving image with a fixed image. Classical methods remain attractive because they provide explicit control over the deformation model, regularization, and numerical stopping criteria, and because they can be applied without training data. Representative approaches include DARTEL, diffeomorphic demons, large-deformation optimal-control formulations, and modern GPU-accelerated variational solvers [1, 13, 8, 12]. Their main practical limitation is computational cost.

Many diffeomorphic schemes update a velocity field through a sequence of convex, linearized optical-flow problems. A typical quadratic subproblem is

$$\min_{v=(v_x, v_y)} \frac{\mu}{2} \|I_x \odot v_x + I_y \odot v_y + I_t\|_2^2 + \frac{1}{2} \|Lv\|_2^2, \quad (1)$$

where I_x, I_y are spatial image derivatives, I_t is the current temporal or residual derivative, and L is a differential regularization operator. Splitting the data and regularization terms produces inexpensive pixelwise and Fourier-domain subproblems, making ADMM particularly suitable for GPU implementation [12, 10].

The penalty ρ and relaxation α strongly affect convergence. Fixed defaults, residual balancing, and adaptive spectral rules can be effective, but they do not exploit the unusually specific geometry of registration: the data Hessian is block diagonal with rank-one 2×2 blocks, while the regularizer is a scalar elliptic operator applied identically to each velocity component. General quadratic ADMM theory derives optimal parameters for special classes [4, 11, 5]; more recently, Song et al. optimized ρ numerically for general linear quadratic problems and derived an optimal α conditional on ρ , including experiments on diffeomorphic registration [10]. Their registration penalty still requires spectral optimization.

This work asks a narrower question:

Can the registration structure be used to replace iterative spectral optimization by a finite, analytically generated parameter set while retaining a rigorous upper bound for the full variable-coefficient iteration?

Our answer is based on local Fourier analysis (LFA), a classical tool for predicting and optimizing stationary iterations for constant-coefficient differential operators [3, 14]. Standard LFA is exact for translation-invariant periodic problems, while nonstandard variants can remain predictive for jumping and random coefficients [7, 9]. Here, the coefficients that vary are the image-gradient blocks in the data Hessian. We freeze those blocks over patches, derive the exact patchwise ADMM symbol, and then control the discrepancy to the full iteration by a resolvent perturbation argument.

Contributions.

1. We derive an exact reduced Cayley-product representation for ADMM and oADMM applied to the split registration quadratic.
2. We introduce a frozen-patch local spectral model in which the ADMM spectrum is explicit. The worst local factor is attained at finitely many data-curvature and regularizer-frequency corners.
3. We prove a noncommuting perturbation bound

$$\varrho(T_{\rho,\alpha}) \leq U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha}{\rho}(\delta_H + \delta_G),$$

where U_{LFA} is the exact frozen-patch envelope and δ_H, δ_G quantify the data and regularizer surrogate errors.

4. We give a finite-candidate, search-free selector. For standard ADMM, the global minimizer of the certified envelope on a prescribed interval is found from explicit corner intersections and quartic stationary equations. Conditional on the selected ρ , the certified oADMM relaxation is selected from a finite closed-form set.
5. We provide a reproducible experimental program designed to test envelope tightness, parameter optimality, and total runtime on synthetic systems and public two- and three-dimensional registration benchmarks.

The manuscript deliberately leaves experimental values blank. No empirical result is asserted before computation.

2 Related Work

2.1 ADMM parameter selection

The scaled ADMM iteration and practical residual stopping criteria were systematized by Boyd et al. [2]. Ghadimi et al. derived optimal penalties and relaxation parameters for several quadratic classes [4]. Teixeira et al. studied scaling, preconditioning, and optimal ADMM parameters in distributed quadratic programming [11]. Giselsson and Boyd developed tight contraction bounds and metric selection for Douglas–Rachford splitting and ADMM under strong-convexity and smoothness assumptions [5]. Adaptive alternatives include residual balancing and Barzilai–Borwein-type spectral updates [15, 16].

Song et al. [10] is the closest work. It formulates ADMM/oADMM for general linear quadratic problems, optimizes the penalty from the iteration spectrum, and gives a closed-form relaxation parameter for fixed penalty. It includes diffeomorphic image registration as an application. The distinction sought here is not another full-spectrum objective: it is a registration-specific local envelope whose certified minimizer is obtained from finitely many algebraic candidates without gradient descent or repeated ADMM trials.

Table 1: Adversarial comparison with the closest work.

Aspect	Song et al. (2024)	This study
Problem class	General linear quadratic problems	Rank-one registration data term and scalar Sobolev regularizer
Penalty	Numerical one-variable spectral optimization	Finite roots of a certified local envelope
Relaxation	Formula conditional on penalty	Finite piecewise-linear candidates conditional on penalty
Variable coefficients	Full iteration spectrum	Local symbol plus noncommuting norm certificate
Empirical conclusion	Imaging improvements reported	Controlled usefulness gate fails; no real-data claim

2.2 ADMM and variational diffeomorphic registration

DARTEL [1] and diffeomorphic demons [13] are influential optimization-based registration methods. Thorley et al. proposed a fast diffeomorphic method that composes nonstationary velocity updates and solves arbitrary-order regularized convex subproblems with accelerated ADMM on GPUs [12]. Song et al. later used a closely related quadratic registration model to evaluate spectrally optimized ADMM and oADMM [10].

2.3 Local Fourier analysis

Fourier analysis has long been used to estimate convergence factors of stationary iterations for elliptic problems [3]. LFA became a principal tool for multigrid design and parameter optimization [14]. Its exactness under periodic translation-invariant assumptions and extensions to broader settings have been studied theoretically [9]. Nonstandard LFA can accurately predict multigrid behavior for random and discontinuous coefficients [7]. Our use differs in two ways: the iteration is

an ADMM splitting map rather than a smoother, and the variable coefficient is a matrix-valued, rank-one optical-flow Hessian.

3 Registration Model and ADMM Splitting

3.1 Discrete quadratic model

Let the image domain contain n grid points in dimension $d \in \{2, 3\}$. Stack the d velocity components into $v \in \mathbb{R}^{dn}$. At grid point x_i , let

$$q_i = \nabla I(x_i) \in \mathbb{R}^d.$$

The linearized data Hessian is the block-diagonal matrix

$$H = \mu \operatorname{diag}(q_1 q_1^\top, \dots, q_n q_n^\top) \succeq 0. \quad (2)$$

Each block has eigenvalues 0 and $\mu \|q_i\|^2$ in two dimensions, or one nonzero and $d-1$ zero eigenvalues in d dimensions.

We assume a componentwise Sobolev regularizer

$$G = L^\top L = G_0 \otimes I_d, \quad (3)$$

where G_0 is a symmetric positive semidefinite finite-difference or spectral discretization. A small zeroth-order term γI or boundary condition may be included to remove the common nullspace. For a first-order periodic regularizer,

$$g(\xi) = \gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^d \sin^2\left(\frac{\xi_\ell}{2}\right) \quad (4)$$

is the Fourier symbol.

The registration subproblem is

$$\min_v \frac{1}{2} v^\top H v + b^\top v + \frac{1}{2} v^\top G v. \quad (5)$$

Introduce $w = v$:

$$\min_{v,w} f(v) + g(w) := \frac{1}{2} v^\top H v + b^\top v + \frac{1}{2} w^\top G w, \quad v - w = 0. \quad (6)$$

3.2 ADMM and over-relaxation

With scaled dual u , standard ADMM is

$$(H + \rho I) v^{k+1} = -b + \rho(w^k - u^k), \quad (7)$$

$$(G + \rho I) w^{k+1} = \rho(v^{k+1} + u^k), \quad (8)$$

$$u^{k+1} = u^k + v^{k+1} - w^{k+1}. \quad (9)$$

For oADMM, replace v^{k+1} in the w and dual updates by

$$\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha) w^k, \quad 0 < \alpha \leq 2.$$

The linear term b shifts the fixed point but does not affect error propagation.

4 Exact Reduced Iteration

Define the resolvents and Cayley transforms

$$P_H(\rho) = \rho(H + \rho I)^{-1}, \quad C_H(\rho) = (H - \rho I)(H + \rho I)^{-1} = I - 2P_H(\rho), \quad (10)$$

$$P_G(\rho) = \rho(G + \rho I)^{-1}, \quad C_G(\rho) = (G - \rho I)(G + \rho I)^{-1} = I - 2P_G(\rho). \quad (11)$$

Proposition 4.1 (Reduced Cayley iteration). *For the quadratic splitting (6), all nonzero eigenvalues of the standard ADMM error matrix coincide with the eigenvalues of*

$$E_{\rho,1} = \frac{1}{2} (I + C_H(\rho)C_G(\rho)). \quad (12)$$

For oADMM, the corresponding reduced operator is

$$E_{\rho,\alpha} = \left(1 - \frac{\alpha}{2}\right) I + \frac{\alpha}{2} C_H(\rho)C_G(\rho). \quad (13)$$

Consequently,

$$\varrho(T_{\rho,\alpha}) = \varrho(E_{\rho,\alpha}).$$

The proof is given in Appendix A. The product $C_H C_G$ is generally nonnormal because H and G do not commute. This is precisely why a global simultaneous diagonalization is unavailable for registration.

5 Frozen-Patch Local Spectral Envelope

5.1 Patch surrogate

Partition the image grid into patches $\{\Omega_r\}_{r=1}^R$. On patch r , choose a representative image-gradient Hessian

$$H_r = \mu \bar{q}_r \bar{q}_r^\top + \nu I_d, \quad (14)$$

where $\bar{q}_r$ may be a center, mean, or robust medoid gradient and $\nu \geq 0$ is an optional model damping already present in the registration objective. Let its eigenvalues be

$$h_{r,1} \leq \dots \leq h_{r,d}.$$

For the rank-one model, $h_{r,1} = \dots = h_{r,d-1} = \nu$ and $h_{r,d} = \nu + \mu \|\bar{q}_r\|^2$.

Let G_r denote a translation-invariant patch regularizer, implemented with periodic patch boundary conditions or another transform-diagonalizable closure. Its symbol lies in

$$g_r(\xi) \in [g_r^-, g_r^+].$$

Construct the block-direct-sum surrogate

$$\tilde{H} = \bigoplus_r (I_{|\Omega_r|} \otimes H_r), \quad \tilde{G} = \bigoplus_r (G_r \otimes I_d).$$

5.2 Exact patch symbol

Because H_r acts only on velocity components and G_r has scalar spatial symbol, the two commute on a patch.

Theorem 5.1 (Frozen-patch spectrum). *The reduced oADMM surrogate $\tilde{E}_{\rho,\alpha}$ is normal. Its eigenvalues are*

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \frac{\alpha}{2} + \frac{\alpha}{2} \frac{h_{r,j} - \rho}{h_{r,j} + \rho} \frac{g_r(\xi) - \rho}{g_r(\xi) + \rho}. \quad (15)$$

Equivalently, with

$$\theta(h, g; \rho) = \frac{\rho^2 + hg}{(\rho + h)(\rho + g)}, \quad (16)$$

we have

$$e_{r,j}(\xi; \rho, \alpha) = 1 - \alpha + \alpha \theta(h_{r,j}, g_r(\xi); \rho).$$

Therefore the exact frozen-patch envelope is

$$U_{\text{LFA}}(\rho, \alpha) = \max_{r,j,\xi} |e_{r,j}(\xi; \rho, \alpha)|. \quad (17)$$

5.3 Corner reduction

The continuous frequency maximization is unnecessary.

Lemma 5.2 (Corner principle). *For fixed $\rho > 0$,*

$$\frac{\partial \theta}{\partial h} = \frac{\rho(g - \rho)}{(g + \rho)(h + \rho)^2}, \quad (18)$$

$$\frac{\partial \theta}{\partial g} = \frac{\rho(h - \rho)}{(h + \rho)(g + \rho)^2}. \quad (19)$$

Hence θ is coordinatewise monotone on every rectangle $[h^-, h^+] \times [g^-, g^+]$, and its minimum and maximum occur at the four corners. Since $|1 - \alpha + \alpha \theta|$ is convex as a function of θ , its maximum also occurs at a corner.

Let

$$\mathcal{C} = \bigcup_{r,j} \{(h_{r,j}, g_r^-), (h_{r,j}, g_r^+)\}. \quad (20)$$

Then

$$U_{\text{LFA}}(\rho, \alpha) = \max_{(h,g) \in \mathcal{C}} |1 - \alpha + \alpha \theta(h, g; \rho)|. \quad (21)$$

Thus the full local Fourier calculation reduces to a small finite set even in three dimensions.

6 Rigorous Variable-Coefficient Bound

The frozen model is useful only if it controls the full noncommuting iteration. Define

$$\delta_H = \|H - \tilde{H}\|_2, \quad \delta_G = \|G - \tilde{G}\|_2, \quad \delta = \delta_H + \delta_G. \quad (22)$$

For block-diagonal H , δ_H is the maximum patchwise 2-norm of a $d \times d$ block difference and is therefore inexpensive. The regularizer error δ_G can be computed by a few power iterations or bounded by a row-sum/Gershgorin estimate of the patch-boundary stencil discrepancy.

Lemma 6.1 (Cayley Lipschitz bound). *For symmetric positive semidefinite A, B and $\rho > 0$,*

$$\|C_A(\rho) - C_B(\rho)\|_2 \leq \frac{2}{\rho} \|A - B\|_2. \quad (23)$$

Theorem 6.2 (Certified local spectral envelope). *Let $T_{\rho, \alpha}$ be the full oADMM error iteration and let U_{LFA} be defined by (21). Then, for $0 < \alpha \leq 2$,*

$$\varrho(T_{\rho, \alpha}) \leq \mathcal{U}(\rho, \alpha) := U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha \delta}{\rho}. \quad (24)$$

The result requires no commutativity assumption on the full matrices H and G .

The proof is in Appendix B. The bound separates two effects:

- U_{LFA} measures the exact contraction of the frozen local models;
- $\alpha \delta / \rho$ penalizes spatial coefficient variation and patch-boundary approximation.

Larger ρ suppresses the perturbation term, while excessively large ρ worsens the local Cayley balance. This tradeoff creates a nontrivial, certifiable parameter-selection problem.

Corollary 6.3 (Guaranteed convergence). *If the selected parameters satisfy $\mathcal{U}(\rho, \alpha) < 1$, then the full linearized registration ADMM/oADMM iteration converges linearly with asymptotic factor at most $\mathcal{U}(\rho, \alpha)$.*

7 Search-Free Parameter Selection

We use “search-free” to mean that no grid search, gradient descent in (ρ, α) , repeated ADMM trial, or repeated full iteration-matrix eigensolve is performed. The method extracts local curvature endpoints once, generates a finite algebraic candidate set, and evaluates the certified scalar envelope.

7.1 Standard ADMM penalty

For $\alpha = 1$, all local modal factors are nonnegative, so

$$\mathcal{U}_1(\rho) = \max_{c \in \mathcal{C}} \theta_c(\rho) + \frac{\delta}{\rho}, \quad \theta_c(\rho) := \theta(h_c, g_c; \rho). \quad (25)$$

Select a physically meaningful compact interval $\mathcal{I} = [\rho_-, \rho_+]$. For example,

$$\rho_- = \epsilon_{\text{num}}, \quad \rho_+ = \max_{c \in \mathcal{C}} \max(h_c, g_c) + \delta,$$

with a small positive numerical floor.

Two corner branches intersect at explicitly computable points. Clearing denominators in $\theta(h_1, g_1; \rho) = \theta(h_2, g_2; \rho)$ gives

$$\rho (A_{12} \rho^2 + B_{12}) = 0, \quad (26)$$

where

$$A_{12} = (h_2 + g_2) - (h_1 + g_1), \quad (27)$$

$$B_{12} = g_1 g_2 (h_1 - h_2) + h_1 h_2 (g_1 - g_2). \quad (28)$$

Thus a positive intersection is

$$\rho_{12} = \sqrt{-B_{12}/A_{12}}$$

whenever the radicand is positive.

On an interval where corner $c = (h, g)$ is active, stationary points of the corrected branch solve

$$(h + g)\rho^2(\rho^2 - hg) - \delta(\rho + h)^2(\rho + g)^2 = 0, \quad (29)$$

which is a quartic polynomial in ρ .

Theorem 7.1 (Finite-candidate ADMM selector). *Let $\mathcal{S}_\rho$ contain:*

1. *the interval endpoints ρ_-, ρ_+ ;*
2. *all positive pairwise intersection roots (26) in $\mathcal{I}$;*
3. *all positive roots of (29) in $\mathcal{I}$ that lie on an interval where the associated corner is active.*

Then every global minimizer of $\mathcal{U}_1$ on $\mathcal{I}$ belongs to $\mathcal{S}_\rho$. Consequently,

$$\rho_{\text{SF}} = \arg \min_{\rho \in \mathcal{S}_\rho} \mathcal{U}_1(\rho) \quad (30)$$

exactly minimizes the certified standard-ADMM envelope without iterative parameter search.

The proof follows from the piecewise-rational structure of the upper envelope and is given in Appendix C.

7.2 Conditional search-free relaxation

After selecting ρ_{SF} , define

$$a = \min_{c \in \mathcal{C}} \theta_c(\rho_{\text{SF}}), \quad (31)$$

$$b = \max_{c \in \mathcal{C}} \theta_c(\rho_{\text{SF}}), \quad (32)$$

with $0 \leq a \leq b < 1$ for a well-posed local model. The certified oADMM envelope conditional on ρ_{SF} is

$$F(\alpha) = \max\{|1 - \alpha + \alpha a|, |1 - \alpha + \alpha b|\} + \frac{\alpha \delta}{\rho_{\text{SF}}}. \quad (33)$$

This is convex and piecewise linear in α .

Theorem 7.2 (Finite-candidate oADMM relaxation). *Let the admissible interval be $[\alpha_-, \alpha_+] \subset (0, 2]$. A minimizer of (33) belongs to*

$$\mathcal{S}_\alpha = \left\{ \alpha_-, \alpha_+, \frac{1}{1-a}, \frac{1}{1-b}, \frac{2}{2-a-b} \right\} \cap [\alpha_-, \alpha_+]. \quad (34)$$

Therefore

$$\alpha_{\text{SF}} = \arg \min_{\alpha \in \mathcal{S}_\alpha} F(\alpha) \quad (35)$$

conditionally minimizes the certified oADMM envelope at ρ_{SF} .

When $\delta = 0$ and the unconstrained equioscillation point lies in $(0, 2]$, the familiar formula

$$\alpha_0 = \frac{2}{2-a-b}$$

is recovered. The perturbation term may instead favor a smaller relaxation.

7.3 Algorithm summary

Algorithm 1: Search-Free Local Spectral Envelope Tuning

1. Compute image gradients and the exact block data Hessian H .
2. Partition the grid; construct H_r and transform-diagonalizable G_r .
3. Extract $h_{r,j}$ and regularizer endpoints g_r^-, g_r^+ ; form $\mathcal{C}$.
4. Compute or bound δ_H and δ_G .
5. Generate $\mathcal{S}_\rho$ from interval endpoints, corner intersections, and active quartic roots.
6. Evaluate $\mathcal{U}_1$ and select ρ_{SF} .
7. Form $\mathcal{S}_\alpha$ and select α_{SF} by evaluating (33).
8. Run oADMM once with the selected parameters. Optionally report $\mathcal{U}(\rho_{\text{SF}}, \alpha_{\text{SF}})$ as a certificate.

8 Implementation Details

8.1 Patch construction

Candidate patch strategies include fixed cubic patches, multiresolution patches matched to the registration pyramid, and adaptive patches split where gradient magnitude or orientation varies beyond a threshold. A robust representative $\bar{q}_r$ can be obtained from the spatial median of gradients. Because H is block diagonal,

$$\delta_H = \max_i \left\| \mu q_i q_i^\top - H_{r(i)} \right\|_2$$

for the direct block surrogate.

8.2 Regularizer symbols

For periodic or cosine-transform boundary closures, $g_r^\pm$ are known analytically. For an order- p Sobolev regularizer with symbol

$$g(\xi) = \left(\gamma + \frac{4\beta}{h^2} \sum_{\ell=1}^d \sin^2(\xi_\ell/2) \right)^p,$$

the endpoints are

$$g^- = \gamma^p, \quad g^+ = \left(\gamma + \frac{4\beta d}{h^2} \right)^p.$$

The patch-boundary discrepancy δ_G can be reduced using overlapping patches or cosine closures rather than disconnected periodic patches.

8.3 Computational complexity

For rank-one data blocks, each patch contributes only two distinct data eigenvalues in 2D or two groups in 3D. Hence $|\mathcal{C}| = O(Rd)$, pairwise corner intersections cost $O(|\mathcal{C}|^2)$, and quartic roots cost $O(|\mathcal{C}|)$. This overhead is independent of the number of ADMM iterations and is typically negligible relative to a 3D registration solve. No full $2n \times 2n$ or $3n \times 3n$ iteration matrix is formed.

9 Reproducible Experiments and Results

All calculations used IEEE double precision, NumPy 2.2.6 and SciPy 1.15.3. The seed was 20260806. Raw JSON/CSV files, the configuration fixed before the controlled run, and generating scripts accompany the paper. The controlled gate was a median relative spectral-radius gap below 0.10. We used dense eigendecomposition only on systems of at most 50 unknowns. Direct joint optimization was a diagnostic differential-evolution calculation (seeded, population multiplier 5, eight generations, followed by polishing), not a proposed selector.

9.1 Experiment 1: constant-coefficient verification

We used $n \times n$ periodic grids, data curvatures (0.2, 2.3), and first-order regularization with $(\beta, \gamma) = (0.2, 0.1)$. The finite selector returned $(\rho, \alpha) = (0.5435121024, 1.8981520705)$ on every grid.

Table 2: Constant-coefficient verification; times are single-process selection times.

n	Direct radius	Envelope	Selection (ms)
4	0.3531304388434793	0.3531304388434793	5.0
6	0.3531304388434803	0.3531304388434793	4.6
8	0.3531304388434788	0.3531304388434793	4.6

9.2 Experiment 2: controlled variable coefficients

We varied grid size (4 and 5), gradient magnitude contrast (1, 4, and 16), and orientation winding (0, 1, and 3), for 18 preconfigured cases. A single mean-Hessian patch deliberately tests the global certificate under mismatch. Every computed full radius satisfied the theorem. Nevertheless, the correction $\alpha\delta/\rho$ dominated selection. In all cases the minimizer was $(\rho, \alpha) = (100, 10^{-6})$, at the allowed boundaries, and the full radius was numerically one. Relative gaps to direct joint spectral optimization had median 2.904, mean 2.839, and maximum 4.505. Thus Gate 2 failed.

Table 3: Controlled variable-coefficient result and predeclared gate.

Measure	Result	Criterion
Certificate violations	0/18	0
Median relative radius gap	2.904	< 0.10
Gate 2	failed	pass

9.3 Decision-gate consequence

The protocol required stopping before expensive registration if the controlled median gap exceeded 0.10. Accordingly, no public medical dataset was downloaded and no Dice, landmark, Jacobian, GPU-memory, or end-to-end runtime value is reported. This is a scientific limitation, but it prevents a loose certificate from being presented as a useful tuner. The implementation still includes residual/objective stopping rules, finite differences, Hessian construction, transform solves, and deformation-Jacobian tests for future work.

10 Discussion

10.1 What is guaranteed

The analysis provides a rigorous upper bound for the full variable-coefficient ADMM/oADMM iteration relative to an explicitly constructed normal local surrogate. The standard-ADMM penalty exactly minimizes this certified envelope over the prescribed interval through finite candidate evaluation. The relaxation parameter exactly minimizes the certified conditional envelope at the selected penalty. These mathematical statements do not imply that the minimizer is a useful contraction.

10.2 What is not claimed

The selected parameters are not claimed to minimize the exact full spectral radius. The guarantee is minimization of a computable upper envelope. Tightness depends on the surrogate errors δ_H and δ_G . The paper also does not claim that ADMM dominates multigrid, Newton–Krylov, or every learned registration method; comparisons concern a particular split registration solver and total runtime under matched accuracy constraints.

10.3 Novelty after comparison with Song et al.

Song et al. numerically reduce the general LQP spectral objective to penalty optimization and give a relaxation formula, whereas our derivation uses the rank-one registration Hessian, a scalar Sobolev symbol, and a finite algebraic envelope with a noncommuting certificate. That distinction survives at the level of analysis. The practical claim does not: the certified selector is markedly worse than direct spectral optimization in our controlled cases. The present contribution is therefore a limitation result and reproducible warning, not a competitive replacement for Song et al.

10.4 Limitations

The certified envelope may be loose when image gradients vary sharply or when patch decoupling produces a large δ_G . Patch selection itself introduces design choices. The theory concerns each convex linearized subproblem; global convergence of the outer nonlinear diffeomorphic registration scheme requires separate assumptions. Higher-order, anisotropic, incompressible, or spatially varying regularizers require matrix-valued symbols, although the same surrogate-and-perturbation framework remains applicable.

10.5 Extensions

Promising extensions include overlapping Schwarz-type local symbols, adaptive patch refinement driven by the certificate, multi-penalty ADMM with separate data and regularization metrics, and online reuse of local curvature summaries across pyramid levels. The envelope could also initialize a single safeguarded Newton correction without reverting to full gradient-based parameter search.

11 Data and Code Availability

The accompanying repository contains source code, tests, the fixed YAML configuration, environment specifications, raw JSON/CSV outputs, the research log, and exact reproduction commands. No external patient data were used: the decision gate failed on generated fields, so downloading or

processing a medical benchmark was neither necessary nor scientifically justified. All synthetic fields are regenerated from the recorded seed.

12 Conclusion

Local spectral analysis gives correct algebra, exact constant-coefficient factors, and a valid non-commuting perturbation certificate. It does not, with the global operator-norm mismatch used here, give a useful joint ADMM/oADMM tuner: the correction rewards vanishing relaxation and large penalty, so the certified minimizer stagnates. The predeclared usefulness gate failed, and claims about real registration speed or quality are therefore unsupported. Progress requires a substantially sharper structured or a posteriori bound—for example, interface-local estimates or weighted resolvent norms—before costly medical-registration evaluation is justified.

A Proof of the Reduced Cayley Representation

For standard ADMM, the homogeneous updates are

$$v^{k+1} = P_H(w^k - u^k), \quad w^{k+1} = P_G(v^{k+1} + u^k).$$

Eliminating v^{k+1} gives the state iteration

$$\begin{bmatrix} w^{k+1} \\ u^{k+1} \end{bmatrix} = \underbrace{\begin{bmatrix} P_G P_H & P_G(I - P_H) \\ (I - P_G)P_H & (I - P_G)(I - P_H) \end{bmatrix}}_{T_{\rho,1}} \begin{bmatrix} w^k \\ u^k \end{bmatrix}.$$

Factor

$$T_{\rho,1} = \begin{bmatrix} P_G \\ I - P_G \end{bmatrix} \begin{bmatrix} P_H & I - P_H \end{bmatrix}.$$

The nonzero eigenvalues of AB and BA coincide. Hence the nonzero spectrum equals that of

$$P_H P_G + (I - P_H)(I - P_G).$$

Using $P_H = (I - C_H)/2$ and $P_G = (I - C_G)/2$ yields

$$P_H P_G + (I - P_H)(I - P_G) = \frac{1}{2}(I + C_H C_G).$$

The relaxed formula follows by inserting $\hat{v}^{k+1} = \alpha v^{k+1} + (1 - \alpha)w^k$ and collecting the same reduced state, yielding

$$E_{\rho,\alpha} = (1 - \alpha)I + \alpha E_{\rho,1} = \left(1 - \frac{\alpha}{2}\right)I + \frac{\alpha}{2}C_H C_G.$$

B Proof of the Perturbation Bound

For $C_A = I - 2\rho(A + \rho I)^{-1}$,

$$\begin{aligned} C_A - C_B &= 2\rho \left[(B + \rho I)^{-1} - (A + \rho I)^{-1} \right] \\ &= 2\rho(A + \rho I)^{-1}(A - B)(B + \rho I)^{-1}. \end{aligned}$$

Since $A, B \succeq 0$,

$$\|(A + \rho I)^{-1}\|_2 \leq \rho^{-1},$$

which proves Lemma 6.1. Next,

$$\begin{aligned} \|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 &\leq \|(C_H - C_{\tilde{H}}) C_G\|_2 + \|C_{\tilde{H}} (C_G - C_{\tilde{G}})\|_2 \\ &\leq \frac{2}{\rho} (\delta_H + \delta_G), \end{aligned}$$

using $\|C_A\|_2 \leq 1$ for $A \succeq 0$. Therefore

$$\begin{aligned} \varrho(E_{\rho, \alpha}) &\leq \|E_{\rho, \alpha}\|_2 \\ &\leq \|\tilde{E}_{\rho, \alpha}\|_2 + \frac{\alpha}{2} \|C_H C_G - C_{\tilde{H}} C_{\tilde{G}}\|_2 \\ &\leq U_{\text{LFA}}(\rho, \alpha) + \frac{\alpha}{\rho} (\delta_H + \delta_G). \end{aligned}$$

C Proof of the Finite-Candidate Selector

Each corner branch θ_c is rational and smooth for $\rho > 0$. The pointwise maximum can change active branch only where two branches intersect, which gives (26). Between consecutive intersections, one branch is active. On such an interval, differentiating

$$\theta(h, g; \rho) + \delta/\rho$$

gives

$$\frac{(h+g)(\rho^2 - hg)}{(\rho+h)^2(\rho+g)^2} - \frac{\delta}{\rho^2}.$$

Clearing positive denominators yields (29). A global minimum of a continuous piecewise-smooth function on a compact interval occurs at an interval endpoint, a nonsmooth branch intersection, or a stationary point of an active smooth branch. This proves Theorem 7.1.

For Theorem 7.2, note that the image of $\theta \in [a, b]$ under $1 - \alpha + \alpha\theta$ is an interval. The maximum absolute value occurs at its endpoints. The resulting maximum of two absolute affine functions plus a linear function is convex piecewise linear. Its slope changes only at the two zero crossings, the equioscillation intersection, and the admissible interval endpoints, proving (34).

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